

UNIFYING THREE-SLOT SLICES: ASSOCIATOR RECOVERY, INTEGRAL CUBES, INFINITE RETURN, AND MARKED JORDAN ARITHMETIC

RESEARCH CONTINUATION FOR THE CLANKERS

ABSTRACT. A three-coordinate vector algebra and a genuinely three-slot tensor are different objects. We retain the distinction and construct explicit maps between the source associator, three-slot cubes, graded cubic-norm operators, octonionic coordinates, and marked Erdős–Straus arithmetic. Three marked pairwise faces leave an exact eight-dimensional joint fibre. A concrete integral $3 \times 3 \times 3$ cube has three identical smooth determinant cubics but an infinite-order round trip through their kernel maps. First-Tits coordinates give three-graded transformations mixing all blocks and changing the quadratic trace while preserving cubic norm. All 196560 minimal vectors in Wilson’s octonionic Leech construction are replayed, including the distribution of their retained ternary Jordan determinant. A marked Erdős–Straus solution is embedded reversibly in an integral cube and then in the infinite-return example, retaining every change of basis and scale. Terzo’s exponential-ring quotient supplies a separate coefficient/period layer; its unconditional period subgroup is distinguished from the complete kernel identified under Schanuel’s conjecture. No universal Erdős–Straus existence theorem, Schanuel proof, or historical-priority claim is made.

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1. SCOPE, CONVENTIONS, AND WHAT IS INHERITED

The requested object is treated as a three-slot construction, not as a claim that writing a real vector space as $\mathbb{R} \oplus \mathbb{C}$ has settled its multiplication, monodromy, or integral arithmetic. Three slots, three spectral values, real dimension, algebraic degree, and coefficient transcendence degree remain different marked quantities. No slice is intrinsically designated “the real part.”

The supplied archive **New folder.zip** contains an explicit three-input associator programme and a Terzo exponential-ring programme. The relevant local claims are read from the supplied mathematical notes and reconstructed below; private conversations and the original thesis PDF

are not redistributed. The source ledger records exactly which mathematical documents were inspected.

The general genus-one interpretation of cubes is inherited from Bhargava and Ho [1, §§3.2.2–3.2.3, 5.1.1]; the first-Tits and exceptional grading framework is classical, with the explicit coordinate presentation in [2, §§5–6 and Appendix A]. Wilson’s Leech construction is imported at its stated scale [4, §§2–4]. The numerical cubes, marked attachments, and joint certificate calculations here are explicitly derived; they are not advertised as historically new theorems. Primary-source searching identified these antecedents, not a comprehensive priority classification.

All matrices act on column vectors. Tensor slots are numbered 0, 1, 2; entries inside a displayed 3×3 matrix are numbered 1, 2, 3 unless stated otherwise. Projective rational vectors are represented in certificates by primitive integer triples whose first nonzero entry is positive. Every inverse below includes its exceptional locus. A retained inverse does not imply that an arbitrary projection is injective.

2. THE SOURCE’S TERNARY OPERATION, BEFORE CONTRACTION

Let k have characteristic different from two, let V have a symmetric bilinear form B , and define the spin product on $k \oplus V$ by

$$(a, u)(b, v) = (ab + B(u, v), av + bu).$$

Use the associator convention $[x, y, z] = (xy)z - x(yz)$.

Proposition 2.1 (Exact ternary recovery). *For $x = (a, u), y = (b, v), z = (c, w)$,*

$$[x, y, z] = (0, B(u, v)w - B(v, w)u).$$

If $u \neq 0$ and $B(u, u) \neq 0$, the operator

$$\mathcal{R}_u(z) = \frac{[(0, u), (0, u), z]}{B(u, u)} = \left(0, w - \frac{B(u, w)}{B(u, u)}u\right)$$

is a projection and

$$\text{Fix}(\mathcal{R}_u) = \text{im}(\mathcal{R}_u) = \text{Ann}(L_{(0, u)}) = \{(0, w) : B(u, w) = 0\}.$$

The nonisotropy condition is the precise domain of this normalization.

Proof. Expanding both parenthesizations cancels their scalar components and all scalar multiples of u, v, w except the two displayed terms. Multiplication by $(0, u)$ is $(c, w) \mapsto (B(u, w), cu)$. Its kernel therefore has $c = 0$ and $B(u, w) = 0$. The displayed operator kills the scalar part and the u component and is the identity on that kernel. This proves idempotence and both equalities. $\square$

For the archive’s normalized blade $J = \langle 1, u, v \rangle$, the form is $B = -\langle \ , \ \rangle$, so $u^2 = v^2 = -1$, $uv = 0$, and its complete associator has four nonzero basis cells:

$$[u, u, v] = -v, \quad [u, v, v] = u, \quad [v, u, u] = v, \quad [v, v, u] = -u.$$

In particular, $-[u, u, v] = v$ and $-[v, v, u] = u$. These statements already occur in the supplied ternary-recovery source; the continuation is not credited with originating them. A contraction using only $\alpha(v, u, -)$ and $\alpha(u, v, -)$ is a noninjective linear map on the space of all trilinear tensors. For example, a tensor supported only on (u, u, v) is invisible to it. Extra knowledge of the full multiplication can of course recover information that a contraction alone has discarded. The source associator is vector-valued: its 27 input cells are vectors, not 27 scalars. Its exact scalar interface is the whole linear family $\ell \mapsto \ell \circ \alpha$ for $\ell \in J^*$. For a basis b_r with dual b_r^* the inverse is $\alpha = \sum_r b_r(b_r^* \circ \alpha)$. Retaining the whole family is lossless; retaining only one scalar component is a further projection and is not called an isomorphism here.

The original unnormalized seed calculation is also independently replayed. With Cayley–Dickson recursion

$$(a, b)(c, d) = (ac - \bar{d}\bar{b}, da + b\bar{c}),$$

left-associated blade words, and C_2 tags, the archive’s seeds in the 32-dimensional tagged sedenion space are

$$X = o_1[0] - o_{1234}[1], \quad Y = o_2[0] + o_{34}[1].$$

The verifier reconstructs the entire basis multiplication, verifies $XY = YX = 0$, $X^2 = Y^2 = -2$, computes both annihilator bases by rational row reduction, and verifies

$$\text{Ann}(L_X) = \text{Fix}(-\tfrac{1}{2}[X, X, -]), \quad \text{Ann}(L_Y) = \text{Fix}(-\tfrac{1}{2}[Y, Y, -]).$$

Both spaces have dimension eight in each case; both containments, not only one, are checked on exact bases. This is a finite linear-algebra certificate for the displayed seeds, not a universal claim about all sedenion zero divisors.

2.1. The local Möbius return is an annihilator line. Take $V = \mathbb{R}^2$ and B negative definite. For

$$u_\theta = (\cos(\theta/2), \sin(\theta/2)), \quad v_\theta = (-\sin(\theta/2), \cos(\theta/2)),$$

Proposition 2.1 projects onto $\mathbb{R}v_\theta$. This line and its projection return at $\theta + 2\pi$, while $v_{\theta+2\pi} = -v_\theta$. Trivializing over $[0, 2\pi]$ therefore identifies $(2\pi, a)$ with $(0, -a)$: the annihilator lines form a Möbius line bundle. A nowhere-zero section would be a continuous nonzero real coefficient with opposite endpoint signs, which is impossible by the intermediate value theorem. This is an actual bundle calculation from the ternary operator. It keeps its local sign return separate from the all-integer return constructed in Section 4.

3. THREE MARKED FACES DO NOT DETERMINE A CUBE

Let V_i be three rank-three free modules over a commutative ring, with marked primitive vectors e_i and functionals ℓ_i satisfying $\ell_i(e_i) = 1$. A cube means a trilinear form

$$T : V_0 \times V_1 \times V_2 \longrightarrow k.$$

Its three marked faces are $F_0 = T(e_0, -, -)$, $F_1 = T(-, e_1, -)$, $F_2 = T(-, -, e_2)$. This is not the collection of all nine coordinate slices.

Theorem 3.1 (Complete three-face gluing fibre). *Faces F_i come from a common T if and only if their common edges agree:*

$$\begin{aligned} g_{01}(w) &= F_0(e_1, w) = F_1(e_0, w), \\ g_{02}(v) &= F_0(v, e_2) = F_2(e_0, v), \\ g_{12}(u) &= F_1(u, e_2) = F_2(u, e_1). \end{aligned}$$

Their common value at all three marks is denoted c . A canonical lift is

$$\begin{aligned} T_0(u, v, w) &= \ell_0(u)F_0(v, w) + \ell_1(v)F_1(u, w) + \ell_2(w)F_2(u, v) \\ &\quad - \ell_0(u)\ell_1(v)g_{01}(w) - \ell_0(u)\ell_2(w)g_{02}(v) \\ &\quad - \ell_1(v)\ell_2(w)g_{12}(u) + c\ell_0(u)\ell_1(v)\ell_2(w). \end{aligned} \tag{1}$$

The complete fibre is

$$T_0 + W_0 \otimes W_1 \otimes W_2, \quad W_i = \{f \in V_i^* : f(e_i) = 0\}.$$

Over a field the face restriction has rank 19 and its kernel has dimension $2 \cdot 2 \cdot 2 = 8$. Over $\mathbb{Z}$ with these primitive marks the same statement is an exact split statement of free abelian groups, with no unmentioned rational lifting condition.

Proof. Necessity follows by evaluating the same trilinear form in two orders. Substituting $u = e_0$ in (1) cancels all terms except F_0 ; substitution in either other slot does the same. For the kernel, use $V_i = ke_i \oplus \ker \ell_i$. A trilinear form vanishing when any slot is its mark is exactly a form on the product of the three complementary modules. Their dual ranks are two. This also gives the inverse: subtract the canonical lift and restrict to the three complements. All steps are integral when the chosen splittings are integral. $\square$

Thus there are eight genuinely joint coefficients invisible to these three marked pairwise views. This eight is not automatically one of the octonionic Peirce spaces: equal dimensions do not supply a map between their different roles. The explicit exceptional-algebra map is given later.

4. AN INTEGRAL CUBE WITH INFINITE THREE-SLICE RETURN

For the following cube, every slot is a copy of $\mathbb{Q}^3$, with ordered bases kept separate. Its first pencil is

$$M_0(x, y, z) = \begin{pmatrix} x & 2z & 3y \\ 3z & y & 2x \\ 2y & 3x & z \end{pmatrix}.$$

Equivalently, its three first-axis arrays are

$$T_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 3 & 0 \end{pmatrix}, \quad T_1 = \begin{pmatrix} 0 & 0 & 3 \\ 0 & 1 & 0 \\ 2 & 0 & 0 \end{pmatrix}, \quad T_2 = \begin{pmatrix} 0 & 2 & 0 \\ 3 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (2)$$

For any slot i , contract that slot and put the remaining two slots, in increasing order, on rows and columns. Direct determinant expansion gives, for each of the three pencils,

$$\det M_i(x, y, z) = -6F(x, y, z), \quad F = x^3 + y^3 + z^3 - 6xyz. \quad (3)$$

The verifier checks all three identities as formal polynomials.

Lemma 4.1 (Smoothness and exact kernel maps). *The three determinant curves C_i over $\mathbb{Q}$ are smooth genus-one curves. At every point of C_i , the contracted matrix has rank exactly two. For $i \neq j$, the incidence condition*

$$T(v_i, v_j, -) = 0$$

with the slots put in their original order defines an isomorphism $\phi_{ij} : C_i \rightarrow C_j$. Its inverse is ϕ_{ji} . A local formula is any nonzero appropriate row or column of the adjugate; a zero chosen cofactor only requires changing that chart.

Proof. A singular point would satisfy $x^2 = 2yz$, $y^2 = 2xz$, $z^2 = 2xy$. A zero coordinate forces all coordinates zero. Otherwise multiplication gives $1 = 8$, a contradiction in characteristic zero. A smooth plane cubic has genus one. If a contracted matrix on the curve had rank at most one, all its cofactors would vanish; differentiation of its determinant would make the point singular, contrary to what was just proved.

A rank-two matrix has a unique projective left kernel and a unique projective right kernel. Use the left one for a row slot and the right one for a column slot. The resulting vector in slot j gives a nonzero kernel vector in the corresponding pencil M_j , so it lies on C_j . Its pencil is again rank two, and the reverse incidence recovers the original point uniquely. Adjugate charts are regular and agree on overlaps because they span the same unique kernel line. This proves the global inverse statement. This is the explicit incidence construction of [1, §§3.2.2, 5.1.1]. $\square$

Keep the full round trip

$$H = \phi_{20}\phi_{12}\phi_{01} : C_0 \longrightarrow C_0.$$

These maps are not to be identified merely because the three curve equations in (3) look identical.

Theorem 4.2 (Faithful all-integer return). *For the cube (2), H has infinite order. In particular, $n \mapsto H^n$ is an injective homomorphism $\mathbb{Z} \rightarrow \text{Aut}(C_0)$.*

Proof. Take $O = [1 : -1 : 0]$ as origin. The triangle-of-isomorphisms theorem [1, §3.2.3, p. 19; §5.1.1 following Lemma 5.3, p. 35] identifies H as translation by a rational point P on the elliptic curve (C_0, O) . Its hypotheses hold: the base field is $\mathbb{Q}$ and the cube is nondegenerate by Lemma 4.1. The direction of the three kernel maps is the direction in that theorem; reversing all maps gives $-P$ and does not change order.

The following exact incidence chain is obtained by row/column cross products:

$$(1, -1, 0) \xrightarrow{0 \rightarrow 1} (2, 3, 1) \xrightarrow{1 \rightarrow 2} (21, -52, 19) \xrightarrow{2 \rightarrow 0} (1817, 5275, 3258).$$

Nine steps give

$$\begin{aligned} H^3(O) = & (666161010410184261713443501009747, \\ & 214674741825814609600406026499153, \\ & 518752964649250744517842481308050), \end{aligned}$$

which is not O . All intermediate points and inverse tests are in the JSON.

The primitive cubic F has smooth reduction at 5 and 17: the same gradient argument works since these primes divide neither 3 nor 7. Exhaustive projective counts, with the first nonzero coordinate set to one, give

$$\#C(\mathbb{F}_5) = 9, \quad \#C(\mathbb{F}_{17}) = 21.$$

The lists of all points are included. Good-reduction injectivity on torsion prime to the residue characteristic follows from [3, Chapter II, Theorem 4.1 and Corollary 4.2, pp. 62–64]. For completeness, the reduction kernel has a separated filtration with successive groups isomorphic to the additive residue field. An element of order prime to that characteristic cannot have a nonzero first image in this filtration. Hence it is zero. Applying this at both 5 and 17 shows that the order of any rational torsion point divides $\gcd(9, 21) = 3$: the 5-primary and 17-primary parts are also excluded by reduction at the other prime. If P were torsion, $3P = 0$, contradicting $H^3(O) \neq O$. Thus P and H have infinite order. This argument is not extrapolation from a finite orbit. $\square$

The result is a three-slot tensor with identical plane-cubic shadows and a retained return invisible to identifying those shadows. The return is stronger than a parity sign: it distinguishes every integer.

5. THREE-GRADED OPERATORS AND THE FULL ALBERT COORDINATES

Take three oriented three-dimensional spaces V_0, V_1, V_2 . For a typed connection to a trilinear cube on input spaces W_0, W_1, W_2 , take $V_i = W_i^*$ and retain the dual bases and volume forms. Then the trilinear coefficient tensor is literally in $V_0 \otimes V_1 \otimes V_2$, not identified with its dual by an unrecorded inner product. Let

$$\mathcal{J} = \text{Hom}(V_0, V_1) \oplus \text{Hom}(V_1, V_2) \oplus \text{Hom}(V_2, V_0).$$

Write its elements as (A, B, C) with the displayed forward arrows. Bases give three 3×3 blocks, hence 27 scalar entries, and define

$$N(A, B, C) = \det A + \det B + \det C - \text{tr}(CBA). \quad (4)$$

The mixed term traverses the complete directed triangle. It cannot be evaluated from three unrelated determinant values. The ordering in (4) is forced by column-vector composition.

This is the first-Tits cubic norm, with the standard block order changed by $(A_0, A_1, A_2) = (A, C, B)$. The basepoint $(I, 0, 0)$ needed to recover a Jordan product is an extra marking; it does not assert an intrinsic “real block.” The associated trace and adjoint in this order are

$$\text{Tr}(A, B, C) = \text{tr } A,$$

$$(A, B, C)^\# = (A^\# - CB, C^\# - BA, B^\# - AC).$$

With $x\#y = (x + y)^\# - x^\# - y^\#$ and $T(x, y) = \text{Tr } x \text{Tr } y - \text{Tr}(x\#y)$, the product is

$$x \circ y = \frac{1}{2}\{x\#y + \text{Tr}(x)y + \text{Tr}(y)x - (\text{Tr}(x)\text{Tr}(y) - T(x, y))1\}.$$

The classical first-Tits theorem supplies its Jordan-algebra status; the coordinate presentation and exact norm/adjoint conventions are [2, §5, equations (10)–(15)].

5.1. A cube acts on the three blocks. For a pure tensor $u = a \otimes b \otimes c \in V_0 \otimes V_1 \otimes V_2$, use oriented cross products and define

$$D_u(A, B, C) = (b(a \times Cc)^T, c(b \times Aa)^T, a(c \times Bb)^T). \quad (5)$$

Extend linearly in all 27 tensor entries. Thus a three-slot cube and a three-block state have different roles: there is an explicit action $T \mapsto D_T \in \text{End}(\mathcal{J})$, rather than an identification based only on equal dimensions. This action is injective with an explicit coefficient inverse. Write $T = \sum t_{ijk} e_i \otimes e_j \otimes e_k$ and let ϵ_{nim} be the orientation sign. For any distinct n, i, m ,

$$t_{ijk} = \epsilon_{nim} (D_T(0, 0, E_{mk})_A)_{jn}. \quad (6)$$

Indeed the displayed matrix entry of the first component of D_T is $\sum_l t_{ljk} \epsilon_{nlm}$, and the distinct-index condition leaves only $l = i$. This proves the inverse over $\mathbb{Z}$ as well as over fields. The checker verifies all 27 coefficient identities as formal linear polynomials.

Theorem 5.1 (Integral simultaneous three-block moves). *For pure u , one has $D_u^2 = 0$, and*

$$U_u(t) = I + tD_u, \quad U_u(t)^{-1} = I - tD_u, \quad N(U_u(t)X) = N(X).$$

Over $\mathbb{Z}$, integral a, b, c, t give an integral automorphism of the 27-entry lattice. In general these maps do not fix the basepoint or preserve the quadratic trace.

Proof. Applying (5) twice produces zero: for example $\delta C c = a(c \times Bb)^T c = 0$, and the other two components are cyclically the same. This proves the inverse.

The coordinate-change group $\mathrm{SL}(V_0) \times \mathrm{SL}(V_1) \times \mathrm{SL}(V_2)$ acts by

$$A \mapsto g_1 A g_0^{-1}, \quad B \mapsto g_2 B g_1^{-1}, \quad C \mapsto g_0 C g_2^{-1}.$$

It preserves (4), and the defining volume-form expression for (5) shows equivariance. When all three vectors are nonzero, volume-one basis changes reduce them to the first basis vector. When one is zero the operator is zero. It suffices over $\mathbb{Q}$, therefore, to check that basis case; the resulting integer polynomial identity then holds over every commutative coefficient ring.

In that case the only changes are

$$\begin{aligned} A_{12} &\mapsto A_{12} - tC_{31}, & A_{13} &\mapsto A_{13} + tC_{21}, \\ B_{12} &\mapsto B_{12} - tA_{31}, & B_{13} &\mapsto B_{13} + tA_{21}, \\ C_{12} &\mapsto C_{12} - tB_{31}, & C_{13} &\mapsto C_{13} + tB_{21}. \end{aligned}$$

Each determinant is affine in t , since only its first row changes. Each increment is $e_1 r$ with $re_1 = 0$, so products of two increments vanish. Any quadratic trace-product term also vanishes by cycling two increments adjacent under trace. The linear derivative satisfies, by a two-minor expansion,

$$\delta \det A = \mathrm{tr}(C \delta B A), \quad \delta \det B = \mathrm{tr}(\delta C B A), \quad \delta \det C = \mathrm{tr}(C B \delta A).$$

For the first equality, both sides are

$$-C_{31}A_{23}A_{31} + C_{31}A_{21}A_{33} + C_{21}A_{21}A_{32} - C_{21}A_{22}A_{31}.$$

The other equalities follow by cycling A, B, C . Thus the entire norm difference is zero. All coefficients in the inverse and transformation are integral. $\square$

For arbitrary T , D_T^2 need not be zero. One must not extend the truncated exponential formula from pure tensors to their sums. Products of pure moves are the exact finite, invertible transformations used here.

The action lies in the classical $\mathbb{Z}/3$ -graded exceptional algebra

$$\mathfrak{g}_0 = \mathfrak{sl}(V_0) \oplus \mathfrak{sl}(V_1) \oplus \mathfrak{sl}(V_2), \quad \mathfrak{g}_1 = V_0 \otimes V_1 \otimes V_2, \quad \mathfrak{g}_2 = V_0^* \otimes V_1^* \otimes V_2^*,$$

with dimensions 24, 27, 27; the resulting algebra is the classical $\mathfrak{e}_6$ representation [2, §6, equations (16)–(20)]. Independently of that identification, `graded_lie.py` constructs the 27 degree-one matrices, extracts 27 independent degree-two commutators, constructs all 24 trace-free degree-zero matrices, and checks all 3003 unordered basis-pair commutators in their prescribed grade. Mixed commutators span all 24 degree-zero directions. Jacobi follows from matrix commutators. Dimension alone is not used as a recognition theorem.

5.2. A spectrum-changing calculation. Let

$$A = \mathrm{diag}(2, 3, 5), \quad B = 0, \quad C = E_{31}, \quad (a, b, c) = (e_2, e_1, e_1).$$

Then the move in Theorem 5.1 gives

$$A(t) = \mathrm{diag}(2 + t, 3, 5), \quad B(t) = 3tE_{13}, \quad C(t) = E_{31}.$$

Define the quadratic trace coefficient

$$S(A, B, C) = \mathrm{tr}(A^\#) - \mathrm{tr}(CB).$$

Direct substitution gives

$$N = 30, \quad S = 31 + 5t, \quad \text{Tr} = 10 + t.$$

Thus $4N - pS = 120 - p(31 + 5t)$ changes. The earlier obstruction for unit-fixing Jordan automorphisms does not apply to these larger transformations. Setting $t = (120/p - 31)/5$ makes this intrinsic invariant equation zero, but this rational split-Albert point has not thereby acquired positive integral spectral coordinates. The equality is recorded at its actual scope, without treating a matrix-state solution as an integer-denominator solution.

5.3. Explicit return to all three octonionic entries. Use Zorn coordinates $q = (\alpha, u, v, \beta)$ with multiplication

$$\begin{aligned} (\alpha, u, v, \beta)(\gamma, w, r, \delta) = & (\alpha\gamma + u \cdot r, \alpha w + \delta u - v \times r, \\ & \gamma v + \beta r + u \times w, v \cdot w + \beta\delta). \end{aligned}$$

The norm is $n(q) = \alpha\beta - u \cdot v$, conjugation is $q^* = (\beta, -u, -v, \alpha)$, and trace is $\alpha + \beta$.

For $X = (A, B, C)$, put $\xi_i = A_{ii}$ and

$$\begin{aligned} x_1 &= (A_{23}, \text{col}_1 B, -\text{row}_1 C, A_{32}), \\ x_2 &= (A_{31}, \text{col}_2 B, -\text{row}_2 C, A_{13}), \\ x_3 &= (A_{12}, \text{col}_3 B, -\text{row}_3 C, A_{21}). \end{aligned}$$

Then

$$\Phi(X) = \begin{pmatrix} \xi_1 & x_3 & x_2^* \\ x_3^* & \xi_2 & x_1 \\ x_2 & x_1^* & \xi_3 \end{pmatrix}. \quad (7)$$

Its inverse reads the diagonal and six scalar corners into A , the upper three-vectors into columns of B , and the negatives of the lower three-vectors into rows of C . Hence all 27 entries are retained, with singleton fibres.

The cubic norm of the image is

$$\xi_1 \xi_2 \xi_3 - \xi_1 n(x_1) - \xi_2 n(x_2) - \xi_3 n(x_3) + \text{tr}((x_1 x_2) x_3) = N(X).$$

One coordinate proof expands the final term as

$$A_{12} A_{23} A_{31} + A_{21} A_{32} A_{13} - \sum_{i \neq j} A_{ij} (CB)_{ji} + \det C + \det B.$$

The diagonal-norm terms contribute the other determinant terms and $-\sum_i A_{ii} (CB)_{ii}$, proving the identity. The complete 45-monomial identity is also checked symbolically by the standard-library verifier. This is the explicit first-Tits/Zorn map of [2, Theorem 11 and Appendix A], with its block order changed as stated, not a new claim of discovering it.

Over $\mathbb{R}$ these Zorn coordinates describe split octonions, not the division real form. Over $\mathbb{C}$, the compact real form is retained by

$$\sigma(A, B, C) = (A^\dagger, C^\dagger, B^\dagger).$$

Indeed its fixed points have $A = A^\dagger$, $C = B^\dagger$, and the trace quadratic form is $\text{tr}(A^2) + 2 \text{tr}(BB^\dagger) > 0$ off zero. On each Zorn entry, the corresponding conjugate-linear involution is

$$(\alpha, u, v, \beta) \mapsto (\bar{\beta}, -\bar{v}, -\bar{u}, \bar{\alpha}),$$

whose fixed norm is $|\alpha|^2 + \|u\|^2$. Formula (7) intertwines these real structures. A general pure move need not preserve σ ; that condition is not discarded. There are also non-unit-fixing norm moves preserving the compact real form: in standard block order (A_0, A_1, A_2) , for $S \in \text{SL}_3(\mathbb{C})$,

$$(A_0, A_1, A_2) \mapsto (SA_0 S^\dagger, (S^\dagger)^{-1} A_1, A_2 S^{-1})$$

preserves the norm and the compact reality equations. Determinants and the cyclic trace prove this directly. Integral unimodular real S preserve the corresponding entry lattice, but integer matrix entries still do not imply integer eigenvalues.

6. A RETAINED TERNARY QUANTITY ON THE LEECH LATTICE

Wilson fixes an octonion basis $1 = i_\infty, i_0, \dots, i_6$ with $i_0 i_1 = i_3$ and the cyclic/doubling subscript rules. Let L be the lattice spanned by the 112 vectors $\pm i_t \pm i_u$ with $t \neq u$, and the 128 vectors

$$\frac{1}{2}(\pm 1 \pm i_0 \pm \dots \pm i_6)$$

with an odd number of minus signs. Put $s = (-1 + i_0 + \dots + i_6)/2$. At Wilson's normalization, the Leech lattice consists of triples satisfying

$$x, y, z \in L, \quad x + y, x + z, y + z \in L\bar{s}, \quad x + y + z \in Ls, \quad (8)$$

with squared norm $(n(x) + n(y) + n(z))/2$ [4, §3, p. 3]. Both the conjugation bar and the order of right multiplication are retained. The identification with the Leech lattice is Wilson's theorem, not inferred here merely from dimension 24.

The linear map

$$\iota(x, y, z) = \begin{pmatrix} 0 & z & \bar{y} \\ \bar{z} & 0 & x \\ y & \bar{x} & 0 \end{pmatrix}$$

puts this lattice into the three Peirce spaces of the compact Albert algebra. Its inverse reads the three marked entries. It satisfies

$$\text{Tr}(\iota(v)^2) = 4\|v\|_\Lambda^2, \quad N_J(\iota(v)) = 2 \text{Re}((xy)z). \quad (9)$$

Thus $\iota/2$ is an isometry for the trace norm. The following basis map makes the change from Wilson's multiplication convention to the preceding Zorn convention fully explicit. In compact Zorn coordinates over $\mathbb{C}$, put

$$I = (i, 0, 0, -i), \quad U_j = (0, e_j, -e_j, 0), \quad V_j = (0, ie_j, ie_j, 0).$$

Then the ordered Wilson basis has images

$$(1, i_0, i_1, i_2, i_3, i_4, i_5, i_6) \mapsto (1, U_1, U_2, I, -U_3, -V_2, -V_3, -V_1). \quad (10)$$

For a proof, these eight vectors lie in the compact fixed real form and are real-linearly independent: their coordinates $(\text{Re } \alpha, \text{Im } \alpha, \text{Re } u_j, \text{Im } u_j)$ form a signed permutation basis. Each imaginary image squares to -1 ; the ordered products in the seven cyclic triples $(t, t+1, t+3)$ agree with Wilson's products. The standard-library certificate checks all 64 basis pairs exactly in $\mathbb{Q}(i)$; bilinearity therefore proves the multiplication identity for arbitrary octonions. The inverse reads the displayed real and imaginary coordinates and reverses the signed permutation. Composing this map entrywise with ι and Φ^{-1} puts the same coupled lattice into the three-block coordinates of Section 5, with its compact real structure retained. It is not being claimed that this off-diagonal lattice is closed under the Jordan product or that every Leech isometry preserves the cubic quantity.

Proposition 6.1 (Complete short-shell cubic distribution). *On Wilson's 196560 norm-four vectors, the cubic value $c(v) = 2 \text{Re}((xy)z)$ has the following exact distribution:*

c	0	± 1	± 3	± 4	± 5	± 7
count for each displayed sign	55248	24288	16800	21504	7392	672.

No other value occurs in this shell. In particular, fixing the quadratic norm has not fixed the retained ternary Jordan determinant.

Proof. We use Wilson's complete minimal-vector description [4, §3, p. 4]. For each root λ of L , signed octonion units j, k , and all permutations, the shapes are

$$(2\lambda, 0, 0), \quad (\lambda\bar{s}, \pm(\lambda\bar{s})j, 0), \quad ((\lambda s)j, \pm\lambda k, \pm(\lambda j)k).$$

The parentheses are part of the formula. The independent checker constructs all 240 roots, all required products, and the resulting sets; it obtains the three disjoint cardinalities 720, 11520, and 184320. It tests every vector against all of (8), not only the generating shapes, and verifies its norm and its cubic value with integer arithmetic.

Here are complete executable membership conventions. Represent an octonion by twice its coordinate vector $v \in \mathbb{Z}^8$. Membership in L is: all coordinates have one parity ε , and $\sum v_i \equiv 2\varepsilon \pmod{4}$. The doubled vectors for $s, \bar{s}$ are respectively $(-1, 1, 1, 1, 1, 1, 1, 1)$ and $(-1, -1, -1, -1, -1, -1, -1, -1)$. If $B(v, w)$ is the integral bilinear multiplication table, membership of $v/2$ in $L\bar{s}$ is exactly that $B(v, 2s)/4$ is an integral vector satisfying the L test. Membership in Ls uses $2\bar{s}$ instead. These formulas follow from $n(s) = 2$ and the octonionic inverse property. For doubled coordinates x, y, z , the cubic value is the scalar coordinate of $B(B(x, y), z)/4$. These bounded loops and the specified parity tests exhaust precisely the three displayed sets. Their resulting histogram is the table; the complete standard-library program, examples of each value, and a sorted-vector hash are supplied. This is an exhaustive finite certificate with Wilson's shell identification as its explicit external input. $\square$

7. MARKED ERDŐS-STRAUS DATA IN A GENUINE CUBE

Fix a positive integer p and an ordered positive integer triple $d = (x, y, z)$. Define an integral tensor by

$$\mathcal{T}_{ijk} = 12d_i\delta_{ij}\delta_{ik} - p, \quad i, j, k \in \{1, 2, 3\}. \quad (11)$$

Its trilinear form is

$$12 \sum_i d_i u_i v_i w_i - p \left(\sum_i u_i \right) \left(\sum_i v_i \right) \left(\sum_i w_i \right).$$

Put $e = (1, 1, 1)$. Every contraction at the corresponding mark is

$$M = 12 \operatorname{diag}(x, y, z) - 3p ee^T. \quad (12)$$

Theorem 7.1 (Exact marked cube criterion). *The encoding (11) is injective. Its inverse is $p = -\mathcal{T}_{112}$ and $d_i = (\mathcal{T}_{iii} + p)/12$ on its image. Moreover,*

$$\det M = 432(4xyz - p(xy + xz + yz)).$$

Thus the marked face is singular exactly when $4/p = 1/x + 1/y + 1/z$. At a passing positive triple it has rank exactly two and kernel $\mathbb{Q}(1/x, 1/y, 1/z)$. No global nondegeneracy of the full cube is assumed.

Proof. The inverse reads the off-diagonal constant and then the three diagonal coefficients, proving injectivity. A determinant expansion of the rank-one perturbation gives

$$\det M = 12^3 xyz \left(1 - \frac{p}{4} \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right) \right),$$

which is the asserted factor. The diagonal matrix is invertible and a rank-one change can reduce its rank by at most one. On a passing triple the vector of reciprocals is in the kernel, so the rank is two and that vector spans it. $\square$

Original selector markings remain attached. For $a = (p + R)/4 = hrs$ and $u = hr^2$, use

$$\begin{aligned} E : \quad R \mid pr + s, \quad \kappa &= (pr + s)/R, & (a, y, z) &= (hrs, h\kappa, phr\kappa), \\ M : \quad R \mid r + s, \quad \lambda &= (r + s)/R, & (a, y, z) &= (hrs, phs\lambda, phr\lambda). \end{aligned}$$

For $0 < R < p$ and the retained channel, the divisor inverse is

$$u = \frac{a^2}{Ry - pa} \quad (E), \quad u = \frac{pa^2}{Ry - pa} \quad (M).$$

Let $d_0 = \gcd(a, u)$; then $r = u/d_0$, $s = a/d_0$, $h = d_0/r$. The positive fibre domain $u \mid a^2$ supplies the integrality. The channel, p, R , and raw denominator order are not dropped when using the cube. The verifier checks these maps on four inherited stress witnesses, including both channels at 1201 and a positive middle witness at 2521.

7.1. Why not identify all three arithmetic views? All eight invisible additions in Theorem 3.1 can be added to (11) without changing its three marked faces. This is a valid joint fibre, not a claim of unrestricted dynamics. There is a specific effect of retaining three *identical* faces: when the resulting cube is nondegenerate and M has reciprocal kernel r , incidence forces $e \mapsto r \mapsto e \mapsto r$. Hence $H^2(e) = e$. Since H is a translation, its translation point has order dividing two. This is a consequence of this particular over-identification, not of all three-slot models.

The following construction avoids it by retaining one arithmetic face and the actual two other typed views.

Theorem 7.2 (An ES mark inside an infinite-return cube). *For every ordered positive ES solution, its matrix M in (12) is the marked first face of a rational cube projectively equivalent to (2). After clearing and retaining one scale, the cube is integral. Its three-slice round trip has infinite order. The complete maps, changes of basis, and scale are invertible and retain the ES mark.*

Proof. Let $A = M_0(1, -1, 0)$ for (2). Both A and M have rank two. Gaussian row and column operations give explicitly invertible matrices

$$U_A A V_A = J_2, \quad U_M M V_M = J_2, \quad J_2 = \text{diag}(1, 1, 0).$$

Set $L = U_M^{-1} U_A$ and $R = V_A V_M^{-1}$, so $LAR = M$. Use

$$G_0 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \quad G_1 = L^T, \quad G_2 = R.$$

Then $G_0 e = (1, -1, 0)$ and the pullback

$$T'(u, v, w) = T(G_0 u, G_1 v, G_2 w)$$

has first marked face M . The inverse applies G_i^{-1} in its own slot. Multiply all coefficients by their common denominator b and retain b ; this changes no projective kernel map. In the new coordinates each kernel map is $G_j^{-1} \phi_{ij} G_i$, so their full composition is $G_0^{-1} H G_0$. Theorem 4.2 proves infinite order. Gaussian pivot choices give different recorded coordinates, not a false uniqueness claim; with the deterministic first-nonzero-pivot convention used in the certificate all data are reproducible. $\square$

For the inherited state

$$(p, R, h, r, s, u, \kappa) = (1201, 23, 17, 1, 18, 17, 53), \quad E,$$

with denominators (306, 16218, 1082101), the certificate obtains $b = 9522$. The integer cube's first pencil arrays are

$$\begin{pmatrix} 657018 & -35621802 & 48768510 \\ 47783466 & -2371784922 & -10183692786 \\ -5511721932 & 284224750146 & 507029017411 \end{pmatrix},$$

$$\begin{pmatrix} 0 & 0 & 1971054 \\ 0 & -27363744 & 1668119022 \\ 19044 & 1824803946 & -118273531557 \end{pmatrix},$$

$$\begin{pmatrix} 0 & 1314036 & -85047330 \\ -82091232 & 4217974452 & 8481265998 \\ 5477395122 & -286083861858 & -265144604956 \end{pmatrix}.$$

Its exact first turn is

$$(1, 1, 1) \rightarrow (63653, 1201, 18) \rightarrow (25855837, 487577, 7866) \rightarrow (1817, 8909, 5075).$$

The first image is the primitive ordered reciprocal triple. The last is the conjugate of $H(O)$, not a reset of the initial mark. This construction encodes an existing witness; it is not used circularly as a production theorem for an unknown prime.

8. TETRAHEDRAL ARITHMETIC AND ITS EXACT INTEGRAL RETURN

The ES equation also has a direct relation to a genuine tetrahedral cubic, not only to an A_4 action on frames. Let

$$\mathcal{C} : e_3(a_0, a_1, a_2, a_3) = 0 \subset \mathbb{P}^3.$$

This is Cayley's nodal cubic in reciprocal coordinates [5, §1, p. 1]. Its four coordinate points are singular and are permuted by S_4 . Direct substitution gives

$$e_3(-p, 4x, 4y, 4z) = 16(4xyz - p(xy + xz + yz)). \quad (13)$$

The preferred negative coordinate is an arithmetic marking; permutations forgetting it need not preserve the same prime problem.

On the torus $a_0 a_1 a_2 a_3 \neq 0$, the reciprocal Cremona map

$$[a_i] \mapsto [b_i], \quad b_i = \prod_{j \neq i} a_j,$$

is a bijection between $\mathcal{C}$ and the plane $\sum b_i = 0$ with no zero coordinate. Its inverse is the same formula. Composing twice gives $a_i(\prod_j a_j)^2$, the same projective point. Coordinate hyperplanes are the explicit excluded locus; the four nodes are not treated as ordinary invertible points of this map.

In the positive ES component the plane coordinates can be oriented as $(-S, A, B, C)$ with $A, B, C > 0$ and $S = A + B + C$. Normalizing by $\gcd(A, B, C) = 1$ gives a unique primitive ordered reciprocal triple.

Proposition 8.1 (Complete positive integral return). *Let A, B, C be positive coprime as a triple, $S = A + B + C$, and $L = \text{lcm}(A, B, C)$. Define*

$$p_0 = \frac{4L}{\gcd(4L, S)}.$$

Then $p_0 \geq 2$. For a prime p , the formulas

$$x = \frac{pS}{4A}, \quad y = \frac{pS}{4B}, \quad z = \frac{pS}{4C}$$

are positive integral ES denominators if and only if $p = p_0$. All ordered positive ES solutions are obtained exactly once from their primitive reciprocal triple in this way.

Proof. The reciprocal sum is $4(A + B + C)/(pS) = 4/p$. The three integrality conditions are jointly $4L \mid pS$, equivalently $p_0 \mid p$. Since $S \leq 3L < 4L$, one has $\gcd(4L, S) < 4L$ and $p_0 \geq 2$; primality makes $p_0 \mid p$ equivalent to $p_0 = p$. Conversely any ES solution gives the primitive positive ratio $A : B : C = 1/x : 1/y : 1/z$. Substituting its reciprocal sum recovers exactly the displayed denominators. Uniqueness follows from primitive positive normalization. $\square$

For each prime ℓ , the retained valuation return is the explicit three-slot piecewise-linear law

$$v_\ell(p_0) = \max\{2 \cdot 1_{\ell=2} + \max(v_\ell A, v_\ell B, v_\ell C) - v_\ell(A + B + C), 0\}.$$

Thus the multiplicative valuations have a simultaneous max-plus description, but the valuation of the actual sum remains essential; its arithmetic carries cannot be replaced by independent slot valuations.

Dividing the coordinates in (13) by $4p$ gives $(-1/4, x/p, y/p, z/p)$. This is an exact occurrence of the marked value $-1/4$ inside ES geometry. No identification with a particular Fable/Jacobian counterexample point is inferred without a map between those different constructions.

9. EXPONENTIAL COEFFICIENTS AND THE RETAINED TERZO PERIOD

The uploaded primary thesis by Giuseppina Terzo [6] uses the free exponential ring $F = [X, Y]^E$, evaluated at $(X, Y) = (\pi, i)$. Put

$$I = \langle Y^2 + 1, E(XY) + 1 \rangle_E, \quad A = F/I.$$

Evaluation $\text{ev} : A \rightarrow \mathbb{C}$ is defined unconditionally because the two relations evaluate to zero. Terzo's Theorem 2.4.1 identifies its kernel as zero *assuming Schanuel's conjecture*. The separate

nonprincipality result is Theorem 2.4.2 in §2.4.1. These statements must not be replaced by a casual “one relation” count.

All integral polynomial cube, norm, adjugate, and pure-move formulas above base change to A . Evaluation commutes with them coefficient by coefficient. If $J = \ker(\text{ev})$, then the kernel on the free cube module is exactly J^{27} . This is an unconditional equality, not an assertion that J vanishes. Under Schanuel, Terzo’s theorem makes the evaluation faithful. The Jordan product further requires localization at two as an underlying ring; no canonical extension of exponentiation to this localization is asserted.

This coefficient layer retains exponential terms in π, i , including nested terms, rather than pretending that they span a three-dimensional vector space over $\mathbb{Q}$. Three slots and coefficient transcendence are independent parts of the specification. For example, powers of any transcendental number are $\mathbb{Q}$ -linearly independent, so a finite-dimensional $\mathbb{Q}$ -algebra closed under multiplication cannot contain that number. This elementary fact does not prevent a rank-three or three-slot construction over the exponential coefficient ring; it specifies which notion of linearity is being used.

There is also an actual all-integer link to Section 4. Put $\varpi = 2\bar{X}\bar{Y} \in A$. Then

$$E(\varpi) = E(\bar{X}\bar{Y})^2 = 1, \quad \text{ev}(\varpi) = 2\pi i.$$

The subgroup $\mathbb{Z}\varpi$ is infinite cyclic unconditionally: evaluating $n\varpi = 0$ forces $n = 0$. Therefore

$$\mathbb{Z}\varpi \longrightarrow \langle H \rangle, \quad n\varpi \longmapsto H^n \tag{14}$$

is an explicit isomorphism of marked groups by Theorem 4.2. It also identifies the actual period lattice $2\pi i\mathbb{Z}$ with these retained cube returns. Under Schanuel only, the complete exponential kernel in A equals $\mathbb{Z}\varpi$: evaluate a kernel element into $2\pi i\mathbb{Z}$ and use injectivity. The unconditional marked subgroup map (14) does not depend on that strengthening. It is a group/lattice map, not a map of exponential rings, an analytic zero-localization theorem, or an ES proof.

Quintic algebraic coefficients can similarly be retained by base change to $k[\theta]/(f(\theta))$. This does not identify tensor arity with polynomial degree, or prove a specific link to a proposed quintic core. Neither that core nor a Schanuel-strength transcendental classification is needed for the unconditional cube and graded-operator theorems.

10. EXECUTABLE SCOPE AND THE RESULTING POINTWISE TARGET

The package provides three standard-library programs:

```
python verify.py --out certificate.json
python -O verify.py
python graded_lie.py
python -O graded_lie.py
python leech_ternary.py
python -O leech_ternary.py
```

Normal and optimized outputs are checked for byte equality separately for each program. No removable Python assertion is used. The first program includes formal sparse-polynomial identities, exact rational kernel calculations, complete reduction point sets, a 27-cell change-of-coordinate inverse, and marked ES stress witnesses. The second verifies all 3003 basis commutators in a 78-dimensional three-graded matrix algebra. The third reconstructs and tests all 196560 listed Wilson minimal vectors. Imported mathematical theorems, formal polynomial checks, and finite exhaustive checks retain different scopes. No Lean run or independent reviewer is claimed; a declaration-level Lean plan is supplied.

The principal construction is no longer confined to a frame rotation. It has three typed views of one tensor, nontrivial all-integer return, an action mixing all three exceptional blocks, an explicit coupled Leech sublattice, and a reversible ES arithmetic mark. The norm-preserving moves genuinely change the quadratic trace, so the earlier frame-only invariant is not an obstruction to this enlarged class.

What has not been proved is that these transformations produce, for every required prime, a positive primitive triple whose exact return value in Proposition 8.1 is that prime. In particular,

an intrinsic split-Albert equation, a nonempty complex orbit, or a rational point in a continuous coordinate chart is not silently converted to a point in that positive integral fibre. A precise next theorem is a terminating three-slot integral operation with $p_0 = p$, or an equivalent marked shell-production operation, while preserving all divisor exponents and signs. The present construction provides the explicit maps and invariant that such an operation must change. It neither proves global ES nor gives an obstruction to the whole proposed programme.

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